Project Proposal

AI-Powered Predictive Maintenance for Industrial Equipment

Objective

To develop an AI-driven predictive maintenance system that uses industrial sensor data (such as vibration, temperature, pressure, and humidity) to predict potential equipment failures before they occur. The system aims to support proactive maintenance, reduce unexpected downtime, and improve maintenance planning.

Description

The project covers the complete predictive-maintenance workflow: data collection and preprocessing, exploratory and advanced time-series analysis, feature engineering, machine learning model development, deployment, and continuous monitoring. The team will use a public dataset, such as data from NASA's Prognostics Center, or simulated sensor readings if a suitable real dataset is unavailable. The final prototype will expose predictions through an API and document how the model can be monitored and retrained when new data becomes available.

Tools & Design

- Programming: Python, Jupyter Notebook
- Data processing: Pandas, NumPy, SciPy, Scikit-learn
- Visualization: Matplotlib, Seaborn, Plotly, and optionally Streamlit or Dash
- Machine learning: Random Forest, XGBoost, and LSTM where appropriate
- MLOps and deployment: MLflow, DVC, FastAPI or Flask, Docker; Kubeflow and Prometheus/Grafana as optional tools
- Data sources: NASA Prognostics Center datasets or simulated industrial sensor data

Technologies to Use

- Exploratory data analysis, data cleaning, and missing-value/outlier handling
- Time-series analysis, trend and cyclical-pattern analysis, and rolling statistics
- Feature engineering, including moving averages, rate of change, and degradation indicators
- Dimensionality reduction with PCA when appropriate
- Time-aware train/validation/test splitting and time-series cross-validation
- Model evaluation using Precision, Recall, F1-score, and ROC-AUC when suitable for the target labels
- Experiment tracking, model versioning, API deployment, model/data-drift monitoring, and retraining workflow

Team Members & Responsibilities

1. Malak Hany — Data Engineer

Milestone 1: collect or simulate sensor data; inspect data structure and quality; perform initial EDA; handle missing values and outliers; create basic features such as moving averages; deliver the cleaned dataset and EDA materials.

2. Jana Mohamed — Data Analyst / Feature Engineer

Milestone 2: conduct advanced time-series analysis; investigate trends and cyclical patterns; apply PCA where appropriate; create rate-of-change, degradation, and rolling-statistics features; build visualizations and summarize findings.

3. Habiba Kamal — Machine Learning Engineer

Milestone 3: select suitable models; prepare time-aware data splits; train and validate models; evaluate results using appropriate metrics; tune hyperparameters and compare models; deliver the model code and selected final model.

4. Sama Saed — MLOps Engineer

Milestone 4: set up experiment tracking and model versioning; package the selected model behind an API; establish monitoring for model/data drift; and prepare a retraining workflow for incoming data.

Week 1 — Milestone 1: Data Collection, Exploration, and Preprocessing

- Malak Hany: obtain a suitable dataset from NASA or create simulated vibration, temperature, and pressure readings; document the source and dataset structure.
- Malak Hany: perform initial EDA, visualize distributions and correlations, identify missing values and outliers, and clean the data.
- Malak Hany: create basic features such as moving averages and save the preprocessed dataset.
- Jana Mohamed: review the initial visualizations and data-quality findings to identify questions for advanced analysis.
- Habiba Kamal: review available target labels and candidate input features to check whether the data can support model training.
- Sama Saed: organize the project folders and establish basic data/version tracking where feasible.

Deliverables

- Dataset Exploration Report
- EDA Notebook
- Cleaned and preprocessed dataset

Week 2 — Milestone 2: Advanced Data Analysis and Feature Engineering

- Jana Mohamed: analyze time-series trends and cyclical patterns in the sensor readings.
- Jana Mohamed: apply PCA where appropriate and explain what the principal components indicate; do not treat PCA alone as proof of failure causes.
- Jana Mohamed: engineer advanced features such as rate of change, degradation indicators, and rolling statistics.
- Jana Mohamed: create plots or a dashboard to communicate degradation patterns, anomalies, and relationships among variables.
- Malak Hany: clarify data-quality decisions and support consistency between the cleaned data and engineered features.
- Habiba Kamal: review feature suitability for predictive modeling and flag potential data leakage risks.
- Sama Saed: document feature-generation steps so they can later be reproduced in the pipeline.

Deliverables

- Advanced Analysis Report
- Visualizations or dashboard
- Feature Engineering Summary and feature-ready dataset

Week 3 — Milestone 3: Predictive Maintenance Model Development and Optimization

- Habiba Kamal: define the prediction target and select candidate models, such as Random Forest, XGBoost, or LSTM depending on the dataset and task.
- Habiba Kamal: split the data into training, validation, and test sets in a time-aware way; use time-series cross-validation where suitable.

- Habiba Kamal: train and compare models using appropriate metrics, including Precision, Recall, F1-score, and ROC-AUC when applicable.
- Habiba Kamal: tune model hyperparameters using Grid Search or another suitable search strategy and compare alternatives before selecting the final model.
- Jana Mohamed: provide the advanced-analysis and feature-engineering outputs needed for model training.
- Malak Hany: verify the final data pipeline and resolve data-quality issues discovered during training.
- Sama Saed: track experiments and model versions with MLflow or an equivalent lightweight approach to support the deployment stage.

Deliverables

- Model Evaluation Report
- Python model-training and evaluation code
- Selected final model file and reproducibility notes

Week 4 — Milestone 4: MLOps, Deployment, and Continuous Monitoring

- Sama Saed: set up experiment tracking and model versioning using MLflow; use DVC or similar data versioning if feasible.
- Sama Saed: expose the selected model through a FastAPI or Flask endpoint for prediction requests.
- Sama Saed: document monitoring for input-data quality and model/data drift; define what conditions should trigger investigation or retraining.
- Sama Saed: implement or demonstrate a retraining workflow that can be run when new labeled data becomes available.
- Habiba Kamal: test the deployed API with representative examples and check that its predictions match the model interface.
- Malak Hany: define how new sensor data should be validated before it enters the prediction/retraining pipeline.
- Jana Mohamed: prepare final charts and summaries explaining the advanced analysis, engineered features, and project findings.
- All team members: integrate notebooks, code, reports, and demo materials; document limitations and future improvements.

Deliverables

- Deployed model API or working deployment prototype
- MLOps and deployment report
- Monitoring and retraining setup or documented prototype
- Final project documentation and demo

Expected Outcome

A documented end-to-end predictive-maintenance prototype that converts sensor data into failure-risk predictions, with a reproducible data and feature pipeline, evaluated machine learning model, API-based inference, and a practical plan for monitoring and retraining.